COURSE TITLE : CHEMICAL ENGINEERING
COURSE CODE : 5093
COURSE CATEGORY : E
PERIODS/WEEK : 4
PERIODS/SEMESTER: 52/5
CREDITS : 4

TIME SCHEDULE

Module	Topics	Period
1	The basic units and unit operations in chemical engineering	13
2	Mechanism of heat transfer by conduction , convection, radiation and evaporation	13
3	Mass Transfer – Diffusion, Absorption, Leaching, Extraction, Adsorption and Drying	13
4	Basic principles of distillation	13
TOTAL		**52**

COURSE OUTCOME

Sl. No.	Sub	Student will be able to
1.	1.	To understand the basic units and unit operations in chemical engineering.
2.	2.	To know mechanism of heat transfer by conduction , convection , Radiation and Evaporation
3.	3.	Mass Transfer – Diffusion, Absorption, Leaching, Extraction, Adsorption and Drying.
4.	4.	To understand the basic principles of distillation

Specific outcome:

Module 1:The basic units and unit operations in chemical engineering 1.1.1 Explain the various systems of units 1.1.2 Explain the difference between fundamental and derived units 1.1.3 Explain the concepts of dimensionless groups 1.1.4 Calculate, using chemical formula, the mass, volume, mole relation, normality 1.1.5 Define gm atom, kg atom, gm mole, kg mole 1.1.6 Solve problems using atomic weight, molecular weight and equivalent weight 1.1.7 Solve problems using mass, volume relationship for gaseous substances 1.1.8 Explain density and specific gravity and specific gravity scales 1.1.9 Solve simple problems in density and specific gravity 1.1.10 Explain various unit operations and unit process in the field of chemical engineering
Module 2 : Heat transfer 2.1.1 Define conduction ,Convection, Radiation 2.1.2 Define thermal conductivity and Fourier's law of thermal conduction 2.1.3 Derivation of conduction equation for a plane wall 2.1.4 Solve the simple problems based on Fourier's law and conduction in plane wall 2.1.5 Derive the equation to calculate heat transfer through composite plane wall 2.1.6 Derive the equation to calculate heat transfer through cylindrical wall 2.1.7 Derive the equation to calculate heat transfer through spherical wall 2.1.8 Solve the problems using the equation derived 2.1.9 Explain the mechanism of natural convection 2.1.10 Explain the heat transfer in boiling liquid and regimes of boiling 2.1.11 An elementary idea of black body, gray body, emissivity, absorptivity, radiation laws and Stefan Boltzmann equation , Heat Transfer equipments 2.1.12 An idea of parallel flow, counter current flow and cross flow heat exchangers 2.1.13 Explain the working Shell and Tube Heat Exchanger and Double pipe heat exchanger with diagram 2.1.14 Mention the various types of evaporators and the basis for classification 2.1.15 Explain the working of horizontal, vertical and multiple effect evaporators
Module 3: Mass Transfer. 3.1.1 State law of Conservation of mass 3.1.2 Define diffusion and Fick's law of diffusion 3.1.3 Explain the mechanism of absorption, Condition of equilibrium between gas and liquid, Henry's law 3.1.4 Explain the factors controlling the rate of absorption 3.1.5 Explain the concept of NTU and HTU, derivation and simple problems 3.1.6 Explain the various packing materials and their characteristics

3.1.7 Explain the working of Packed Tower, flooding and channeling in packed columns 3.1.8 Define leaching and its applications– batch and continuous leaching , heap leaching 3.1.9 Explain the operation of Boll man extractor, horizontal and vertical type 3.1.10 Define liquid- liquid extraction, the terms raffinate and extract 3.1.11 Mention the various extraction equipments 3.1.12 Explain the construction details and working of Mixer settler 3.1.13 [Explain adsorption, physical and chemical adsorption and the various types of adsorbents 3.1.14 Mention the purpose and applications of drying 3.1.15 List the various drying equipments 3.1.16 Explain the working of Tray drier and Spray drier
Module 4: Understand the Basic Principles of Distillation 4.1.1 Define Distillation and types of Distillation –Equilibrium, simple, steam and fractional distillation. 4.1.2 Vapour liquid equilibria and phase diagram for a binary mixture 4.1.3 Explain Simple Distillation and Steam Distillation with diagram 4.1.4 Define Raoult's law and derive equation – ideal and non ideal solutions 4.1.5 Define relative volatility and derive the equation for relative volatility 4.1.6 Solve simple problems related to Raoult's law and relative volatility 4.1.7 Describe maximum and minimum boiling Azeotropes 4.1.8 Compare Azeotropic and Extractive Distillation

CONTENT DETAILS

MODULE I

UNITS AND DIMENSIONS

Units and Dimensions-various systems of units-fundamental & derived units-dimensionless groups-mass, volume, mole relation, normality-atomic weight, molecular weight, equivalent weight-mass, volume, relationships for gaseous systems-density, specific gravity-simple problems-concept of unit operations and unit process

MODULE II

HEAT TRANSFER

Conduction, Convection, Radiation - definition - Fourier's law –derivation - problems – Conduction through Plane wall ,Composite wall, Cylindrical wall, Spherical wall – derivations – problems –concept of natural convection –heat transfer in boiling of liquids – idea of black body, gray body, emissivity, absorptivity, radiation laws, Stefan Boltzmann equation– Shell and Tube, Double Pipe Heat Exchangers – diagrams- Evaporators – basis of classification - horizontal, vertical and multiple effect evaporator.

MODULE III

MASS TRANSFER

Diffusion – Fick's rate equation – absorption – gas liquid equilibrium –Hendry's law – concept of NTU and HTU – derivation – problems –operation of packed tower – flooding – channeling – leaching – applications – working of Boll man Extractor – Extraction – raffinate, extract, - working of mixer settler – adsorption – physical and chemical adsorption – types of adsorbents – Drying – applications – list of equipments – working of tray and spray drier

MODULE IV

DISTILLATION

Distillation – types – Simple and Steam distillation –fractional distillation –vapor liquid equilibrium-phase – diagram – Raoult's law – Relative volatility – derivation – simple problems –Minimum and Maximum boiling Azeotropes – comparison - Azeotropic and Extractive Distillation.

REFERENCE

1. Process Calculations for chemical engineers - Chem. Engg., Edn Development Centre, IIT Madras

2. Process Calculations - V. Venkataramani, N. Anantharaman

3. Introduction to Chemical Engineering - Walter.L.Badger (Publisher: McGraw-Hill Inc.,US; International 1955 Edition)

4. Unit Operations - McCABE SMITH(Publisher - McGraw-Hill Education; 7edition)

5. Mass Transfer - Robert E Treybal (Publisher - McGraw-Hill Book Company; 3rd edition)